

JEE MAIN MOCK TEST SAMPLE

Subjects : Physics , Chemistry ,
Maths

JEE MAIN MOCK SAMPLE

Total Marks : 147

Physics - Section A (MCQ.)

1. Two friends A and B are standing a distance x apart in an open field and wind is blowing from A to B . A beats a drum, and B hears the sound t_1 time after seeing the event. A and B swap their positions and the experiment is repeated. This time, B hears the drum t_2 time after seeing the event. Determine the velocity of sound in still air v and the velocity of wind u . Neglect the time light takes to travel between the friends.

A) $v = \frac{2x}{t_1 + t_2}, u = \frac{2x}{t_1 - t_2}$

B) $v = \frac{x}{2} \left(\frac{1}{t_1} + \frac{1}{t_2} \right), u = \frac{x}{2} \left(\frac{1}{t_1} - \frac{1}{t_2} \right)$

C) $v = \frac{x}{2} \left(\frac{1}{t_1} - \frac{1}{t_2} \right), u = \frac{x}{2} \left(\frac{1}{t_1} + \frac{1}{t_2} \right)$

D) $v = x \left(\frac{1}{t_1} + \frac{1}{t_2} \right), u = x \left(\frac{1}{t_1} - \frac{1}{t_2} \right)$

2. An aeroplane is required to travel from a point A to another point B , located 500 km away due 30° east of north. A wind blows due north at a speed of 20 m/s, and the air-speed of the plane is 150 m/s. Determine the direction in which the pilot should head the plane to reach point B .

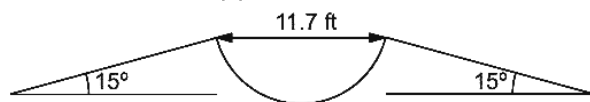
A) $\sin^{-1} \left(\frac{\sqrt{3}}{15} \right)$ east of the line AB

B) $\sin^{-1} \left(\frac{1}{15} \right)$ east of the line AB

C) $\sin^{-1} \left(\frac{1}{15} \right)$ west of the line AB

D) $\sin^{-1} \left(\frac{\sqrt{3}}{15} \right)$ west of the line AB

3. As shown in the figure, a ditch is 11.7 ft wide and has approach roads inclined at an angle of 15° with the horizontal. Determine the minimum speed in ft/s at which a motorbike must travel on the road to safely cross the ditch. Assume the bike has a length of 5 ft and that it leaves the road exactly when its front part runs out of the approach road.



A) 30.0

B) 32.5

C) 35.5

D) 27.5

4. A temperature difference can generate e.m.f. in some materials. Let S be the e.m.f. produced per unit temperature difference between the ends of a wire, σ the electrical conductivity and κ the thermal conductivity of the material of the wire. Taking M, L, T, I and K as dimensions of mass, length, time, current and temperature, respectively, the dimensional formula of the quantity $Z = \frac{S^2 \sigma}{\kappa}$ is :-

A) $[M^0 L^0 T^0 I^0 K^0]$

B) $[M^0 L^0 T^0 I^0 K^{-1}]$

C) $[M^1 L^2 T^{-2} I^{-1} K^{-1}]$

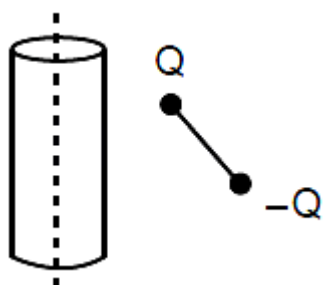
D) $[M^1 L^2 T^{-4} I^{-1} K^{-1}]$

5. In a historical experiment to determine Planck's constant, a metal surface was irradiated with light of different wavelengths. The emitted photoelectron energies were measured by applying a stopping potential. The relevant data for the wavelength (λ) of incident light and the corresponding stopping potential (V_0) are given below:

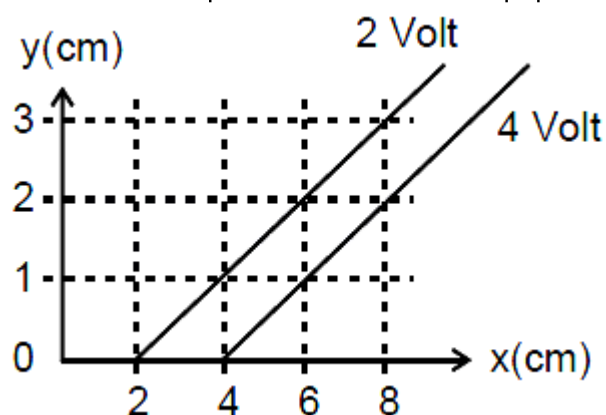
$\lambda(\mu m)$	Volt
0.3	2.0
0.4	1.0
0.5	0.4

Given that $c = 3 \times 10^8 m s^{-1}$ and $e = 1.6 \times 10^{-19} C$, Planck's constant (in units of J s) found from such an experiment is :

- A) 6.0×10^{-34} B) 6.4×10^{-34} C) 6.6×10^{-34} D) 6.8×10^{-34}
6. Consider a positively charged infinite cylinder with uniform volume charge density $r > 0$. An electric dipole consisting of $+Q$ and $-Q$ charges attached to opposite ends of a massless rod is oriented as shown in the figure. At the instant as shown in the figure, the dipole will experience,

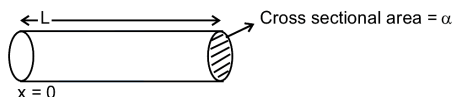


- A) a force to the left and no torque
 B) a force to the right and a clockwise torque
 C) a force to the right and a counter clockwise torque
 D) non force but only a clockwise torque
7. The figure represents two equipotential lines in $x - y$ plane for an electric field. The x -component E_x of the electric field in space between these equipotential lines is,

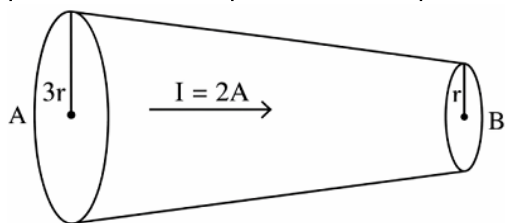


- A) 100 V/m B) -100 V/m C) 200 V/m D) -200 V/m

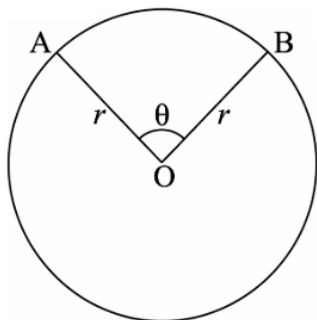
8. The variation of density of a solid cylindrical rod of cross sectional area α and length L is $\rho = \rho_0 \frac{x^2}{L^2}$, where x is the distance from one end of the rod. The position of its centre of mass from one end ($x = 0$) is



- A) $2L/3$ B) $L/2$ C) $L/3$ D) $3L/4$
9. Two spheres S_1 and S_2 of masses m_1 and m_2 respectively collide with each other. Initially S_1 is at rest and S_2 is moving with velocity v along x -axis. After collision S_2 has a velocity $\frac{v}{2}$ in a direction perpendicular to the original direction. The sphere S_1 moves after collision
- A) with a velocity of magnitude $\frac{m_2}{m_1} v \frac{\sqrt{5}}{2}$
- B) with a velocity in the direction $\theta = \tan^{-1} \left(-\frac{1}{3} \right)$ to the x -axis
- C) with a velocity whose direction makes an angle θ with x -axis such that $\theta = \tan^{-1} \left(\frac{1}{2} \right)$ or $\theta = \tan^{-1} \left(-\frac{1}{2} \right)$
- D) with a velocity of magnitude $\frac{m_1}{2m_2} v \sqrt{5}$
10. As following figure 2 A current passing through a conducting wire, radius of cross-sectional of wire at point A is $3r$ and point B is r respectively. Then find the ratio of drift velocity at point A and B.

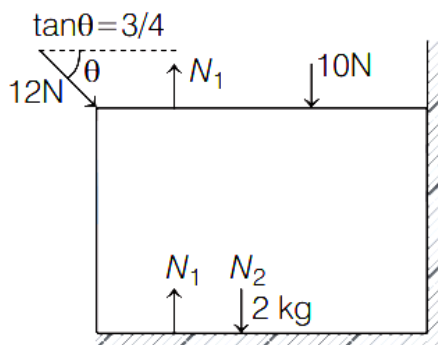


- A) 3 B) $\frac{1}{9}$ C) $\frac{1}{3}$ D) 9
11. A and B are two points on a uniform ring of radius r . The resistance of the ring is R . $\angle AOB = \theta$ as shown in the figure. The equivalent resistance between points A & B is _____.

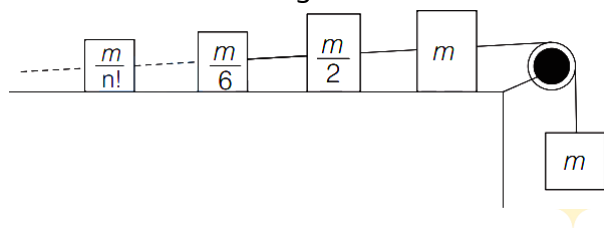


- A) $\frac{R\theta}{2\pi}$ B) $\frac{R(2\pi-\theta)}{4\pi}$
- C) $R \left(1 - \frac{\theta}{2\pi} \right)$ D) $\frac{R}{4\pi^2} (2\pi - \theta) \cdot \theta$
12. If the moment of inertia of a uniform solid cylinder about the axis of the cylinder is $\frac{1}{n}$ times its moment of inertia about an axis passing through its midpoint and perpendicular to its length, then the ratio of the length and radius of the cylinder is
- A) $\sqrt{2(3n+1)}$ B) $\sqrt{2(3n-1)}$ C) $\sqrt{3(2n+1)}$ D) $\sqrt{3(2n-1)}$
13. The potential energy of a particle in a central field has the form $U(r) = \frac{1}{r^2} - \frac{1}{r}$, where ' r ' is the distance from the centre of the field. The magnitude of the maximum attractive force in Newton is
- A) $\frac{1}{27}$ B) $\frac{1}{9}$ C) $\frac{1}{3}$ D) 1

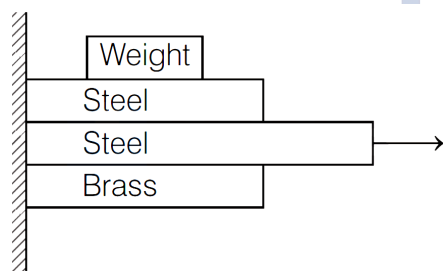
14. A block is between two surfaces as shown in the figure. Find the normal reaction at both surfaces. [Assume, $g = 10 \text{ m/s}^2$]



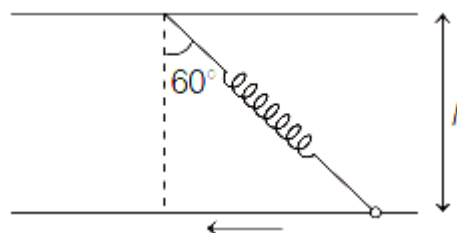
- A) $N_1 = 37.2 \text{ N}$ and $N_2 = 9.6 \text{ N}$ B) $N_1 = 38.2 \text{ N}$ and $N_2 = 8.6 \text{ N}$
 C) $N_1 = 40 \text{ N}$ and $N_2 = 4 \text{ N}$ D) $N_1 = 37.5 \text{ N}$ and $N_2 = 9.9 \text{ N}$
15. An infinite number of masses are placed on a frictionless table and they are connected via massless strings. Their masses follow the sequence, $m, \frac{m}{2}, \frac{m}{6}, \dots, \frac{m}{n!}, \dots$ and they are further connected to a mass m that hangs over a massless pulley. The acceleration of the hanging mass is



- A) $\frac{g}{e-1}$ B) $\frac{g}{e+1}$ C) $\frac{g}{e}$ D) $\frac{g}{2e}$
16. A movable steel plate is placed between fixed steel and brass plates and the stack of plates is subjected to a weight of 100 N as shown in the figure. The coefficient of kinetic friction for steel on steel is 0.57 and for steel on brass is 0.44. Assuming that the entire weight comes onto the stack and that the weight of the plates is negligible in comparison to the applied weight, the force required to move the middle plate (in N) is



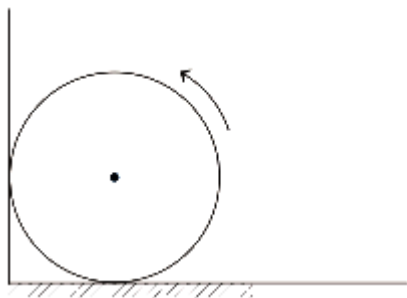
- A) 13 B) 101 C) 440 D) 570
17. A spring has a natural length l with one end fixed to the ceiling. The other end is fitted with a smooth ring which can slide on a horizontal rod fixed at distance l below the ceiling. Initially, the spring makes an angle of 60° with the vertical, when system is released from rest. Find the angle of the spring with the vertical, when the velocity of the ring reaches half of the maximum velocity, which the ring can attain



during the motion.

- A) 30° B) $\cos^{-1} \left(\frac{2}{2+\sqrt{3}} \right)$
 C) $\cos^{-1} \left(\frac{\sqrt{3}-1}{2} \right)$ D) None of the above

18. A uniform cylinder of radius 1 m, mass 1 kg spins about its axis with an angular velocity 20rad/s . At certain moment, the cylinder is placed into a corner as shown in the figure. The coefficient of friction between the horizontal wall and the cylinder is μ , whereas the vertical wall is frictionless. If the number of rounds made by the cylinder is 5 before it stops, then the value of μ is (acceleration due to gravity, $g = 10\text{ m/s}^2$)



- A) $\frac{3}{\pi}$ B) $\frac{2}{\pi}$ C) $\frac{1}{\pi}$ D) $\frac{0.4}{\pi}$
19. Two identical vessels A and B with frictionless pistons contain the same ideal gas at the same temperature and the same volume V . The masses of gas in A and B are m_A and m_B , respectively. The gases are allowed to expand isothermally to the same final volume $2V$. The change in pressures of the gas in A and B are found to be ΔP and $1.5\Delta P$, respectively. Then
- A) $9m_A = 4m_B$ B) $3m_A = 2m_B$ C) $2m_A = 3m_B$ D) $4m_A = 9m_B$
20. One mole of the ideal gas goes through the process $p = p_0 \left[1 - \alpha \left(\frac{V}{V_0} \right)^3 \right]$, where p and V are pressure and volume, p_0 , V_0 and α are constants. If the maximum attainable temperature of the gas is $\left(\frac{3}{4} \right) \frac{p_0 V_0}{R}$, then the value of α is
- A) 2 B) $\frac{1}{2}$ C) $\frac{1}{4}$ D) 4
21. An ideal gas has molar heat capacity C_V at constant volume. The gas undergoes a process where in the temperature changes as $T = T_0 (1 + \alpha V^2)$, where, T and V are temperature and volume respectively, T_0 and α are positive constants. The molar heat capacity C of the gas is given as $C = C_V + Rf(V)$, where, $f(V)$ is a function of volume. The expression for $f(V)$ is
- A) $\frac{\alpha V^2}{1 + \alpha V^2}$ B) $\frac{1 + \alpha V^2}{2\alpha V^2}$
 C) $\alpha V^2 (1 + \alpha V^2)$ D) $\frac{1}{2\alpha V^2 (1 + \alpha V^2)}$
22. Consider a parallel plate capacitor with plates in the shape of square and in XY -plane. The gap between the plates is filled with dielectric material. The dielectric constant k varies with X -axis as $k(x) = \left[1 + \left(\frac{x}{L} \right)^\alpha \right]$, where α is a constant. Let C_d and C_a are capacitance in the presence of dielectric and air, respectively. If the ratio $\frac{C_d}{C_a} = \frac{7}{6}$, then the value of α must be
- A) 3 B) 5 C) 7 D) 9
23. A capacitor of capacity $0.1\mu\text{F}$ connected in series to a resistor of $10\text{M}\Omega$ is charged to a certain potential and then made to discharge through the resistor. The time in which the potential will take to fall to half its original value is (Given, $\log_{10} 2 = 0.3010$)
- A) 2 s B) 0.693 s C) 0.5 s D) 1.0 s
24. The radius of the first orbit of hydrogen is r_H , and the energy in the ground state is -13.6eV . Considering a μ^- -particle with a mass $207m_e$ revolving round a proton as in hydrogen atom, the energy and radius of proton and μ^- -combination respectively in the first orbit are (assume nucleus to be stationary)
- A) $-13.6 \times 207\text{eV}$, $\frac{r_H}{207}$ B) -207
 $\times 13.6\text{eV}$, $207r_H$ C) $-\frac{13.6}{207}\text{eV}$, $\frac{r_H}{207}$ D) $-\frac{13.6}{207}\text{eV}$, $207r_H$
25. Current I through a given $p - n$ junction when a voltage V is applied across it is given to be $I = I_0 \left(e^{\frac{V}{V_T}} - 1 \right)$ where I_0 and V_T are constants. If $r_d(I)$ is the dynamic resistance of the junction, then $r_d(1000I_0) = \alpha r_d(10I_0)$, where α is approximately equal to

Chemistry - Section A (MCQ.)

26. The incorrect postulates of the Dalton's atomic theory are :

- (A) Atoms of different elements differ in mass.
 (B) Matter consists of divisible atoms.
 (C) Compounds are formed when atoms of different element combine in a fixed ratio.
 (D) All the atoms of given element have different properties including mass.
 (E) Chemical reactions involve reorganisation of atoms.

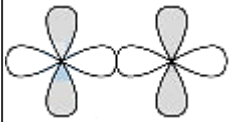
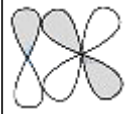
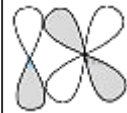
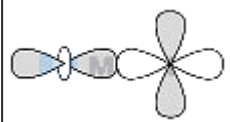
Choose the correct answer from the options given below :

- A) (B), (D), (E) only B) (A), (B), (D) only
 C) (C), (D), (E) only D) (B), (D) only

27. The correct statements (s) about intermolecular forces is(are)

- A) The potential energy between two point charges approaches zero more rapidly than the potential energy between a point dipole and a point charge as the distance between them approaches infinity.
 B) The average potential energy of two rotating polar molecules that are separated by a distance r has $1/r^3$ dependence.
 C) The dipole-induced dipole average interaction energy is independent of temperature.
 D) Nonpolar molecules attract one another even though neither has a permanent dipole moment.

28. Match the orbital overlap figures shown in List-I with the description given in List-II and select the correct answer using the code given below the lists.

	List - I		List - II
A.		P	p - d π antibonding
B.		Q	d - d σ bonding
C.		R	p - d π bonding
D.		S	d - d σ antibonding

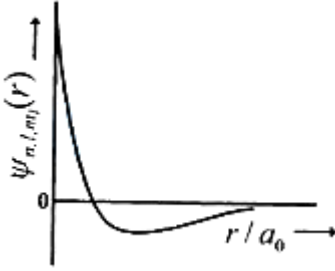
- A) a-q;b-r;c-p;d-s; B) a-r;b-q;c-p;d-s; C) a-r;b-q;c-s;d-p; D) a-p;b-s;c-r;d-q;

29. The stoichiometric reaction of 516 g of dimethyldichlorosilane with water results in a tetrameric cyclic product X in 75% yield. The weight (in g) of X obtained is _____

[Use, molar mass. (gmol^{-1}) : H = 1, C = 12, O = 16, Si = 28, Cl = 35.5]

- A) 111 B) 222 C) 333 D) 444

30. The wave function, ψ_{n, l, m_l} is a mathematical function whose value depends upon spherical polar coordinates (r, θ, ϕ) of the electron and characterized by the quantum numbers n, l and m_l . Here r is the distance from the nucleus, θ is colatitude and ϕ is azimuth. In the mathematical functions given in the Table, Z is the atomic number and a_0 is Bohr radius

Column 1	Column 2	Column 3
(I) 1s orbital	(i) ψ_{n, l, m_l} $\propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$	(P) 
(II) 2s orbital	(ii) One radial node	(Q) Probability density at the nucleus $\propto \frac{1}{a_0^3}$.
(III) $2p_z$ orbital	(iii) ψ_{n, l, m_l} $\propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} r e^{-\left(\frac{Zr}{2a_0}\right)} \cos \theta$	(R) Probability density is maximum at the nucleus.
(IV) $3d_z^2$ orbital	(iv) xy -plane is a nodal plane	(S) Energy needed to excite an electron from $n = 2$ state to $n = 4$ state is $\frac{27}{32}$ times the energy needed to excite an electron from $n = 2$ state to $n = 6$ state.

For the given orbital in column 1, the only correct combination for any hydrogen-like species is

- A) (III) (iii) (P) B) (I) (ii) (S)
C) (IV) (iv) (R) D) (II) (ii) (P)

31. The order of basic strength of methyl substituted amines and NH_3 in aqueous solution is

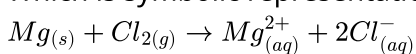
- A) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{NH}_3$ B) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > \text{NH}_3$
C) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > \text{NH}_3 > (\text{CH}_3)_3\text{N}$ D) $\text{NH}_3 > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$

32. Assertion (A): Carboxylic acids are more acidic than Phenols

Reason (R): Resonance structures of carboxylate ion are equivalent, while resonance structures of phenoxide ion are not equivalent

- A) Both (A) and (R) are correct and (R) is the correct explanation of (A)
B) Both (A) and (R) are correct But (R) is not the correct explanation of (A)
C) (A) is correct but (R) is incorrect
D) (A) is incorrect but (R) is correct

33. Which is symbolic representation for following cell reaction?



- A) $\text{Mg} | \text{Mg}_{(aq)}^{2+} (1M) || \text{Cl}_{(aq)}^- (1M) | \text{Cl}_{2(g)} (1\text{bar}) | \text{Pt}$
B) $\text{Pt} | \text{Cl}_{(aq)}^- (1M) | \text{Cl}_{2(g)} (1\text{bar}) || \text{Mg}_{(aq)}^{2+} (1M) | \text{Mg}$
C) $\text{Mg} | \text{Mg}_{(aq)}^{2+} (1M) || \text{Cl}_{2(g)} (1\text{bar}) | \text{Cl}_{(aq)}^- (1M) | \text{Pt}$
D) $\text{Pt} | \text{Cl}_{2(g)} (1\text{bar}) | \text{Cl}_{(aq)}^- (1M) || \text{Mg}_{(aq)}^{2+} (1M) | \text{Mg}$

34. Match List-I with List-II for the following reaction pattern

Glucose $\xrightarrow{\text{Reagent}}$ Product $\rightarrow$ Structural prediction

	List - I (Reagents)		List-II (Structural prediction)
a.	Acetic anhydride	(i)	Glucose has an aldehyde group
b.	Bromine water	(ii)	Glucose has a straight chain of six carbon atoms
c.	Hydroiodic acid	(iii)	Glucose has five hydroxyl group
d.	Hydrogen cyanide	(iv)	Glucose has a carbonyl group

Choose the correct answer from the options given below.

A) a-iv, b-iii, c-ii, d-i

B) a-iii, b-i, c-ii, d-iv

C) a-i, b-ii, c-iii, d-iv

D) a-iii, b-ii, c-i, d-iv

35. The glycosidic linkage involved in linking the glucose units in amylose part of starch is

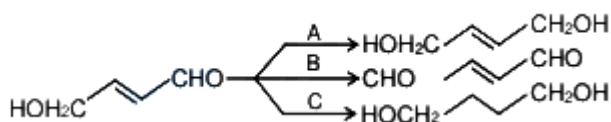
A) $C_1 - C_4\beta$ - linkage

B) $C_1 - C_6\alpha$ - linkage

C) $C_1 - C_6\beta$ - linkage

D) $C_1 - C_4\alpha$ - linkage

36.



The reagents A, B and C respectively are

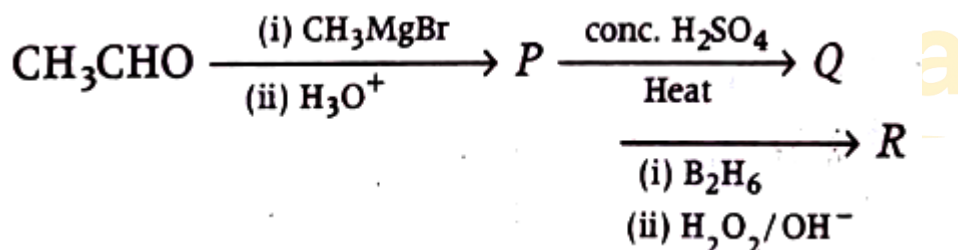
A) NaBH_4 , PCC, H_2/Pd

B) H_2/Pd , alk. KMnO_4 , NaBH_4

C) H_2/Pd , PCC, NaBH_4

D) NaBH_4 , alk. KMnO_4 , H_2/Pd

37. Compounds P and R in the following reaction are



A) Position isomers

B) Functional isomers

C) Metamers

D) Identical

38. When FeCl_3 is added to excess of hot water gives a sol 'X'. When FeCl_3 is added to NaOH(aq) solution, gives sol 'Y'. X and Y formed in the above processes respectively are

A) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}/\text{OH}^-$ and $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}/\text{Fe}^{3+}$

B) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}/\text{H}^+$ and $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}/\text{Na}^+$

C) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}/\text{Cl}^-$ and $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}/\text{OH}^-$

D) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}/\text{Fe}^{3+}$ and $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}/\text{O}^-$

39. The rate constant of a reaction is given by $k = Ae^{-E_a/RT}$ under standard notation. In order to speed up the reaction, which of the following factors has to be decreased?

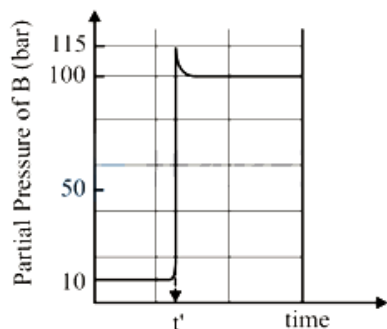
A) Z

B) Both Z and T

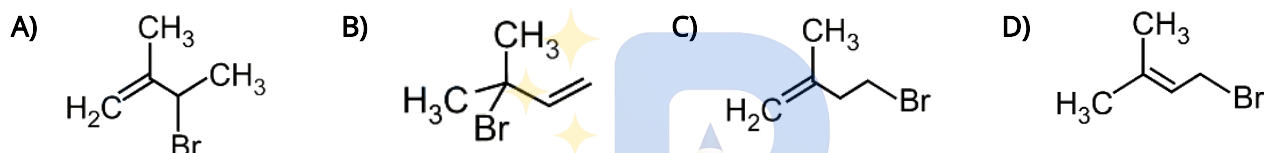
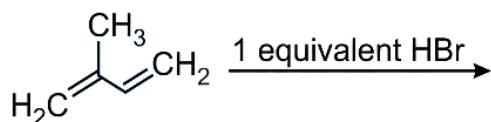
C) E_a

D) T

40. Consider the reaction $A \rightleftharpoons B$ at 1000 K. At time t' , the temperature of the system was increased to 2000 K and the system was allowed to reach equilibrium. Throughout this experiment the partial pressure of A was maintained at 1 bar. Given below is the plot of the partial pressure of B with time. What is the ratio of the standard Gibbs energy of the reaction at 1000 K to that at 2000 K?



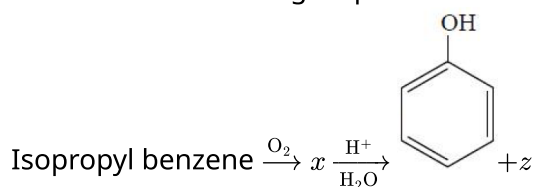
- A) 0.25 B) 0.24 C) 0.3 D) 0.39
41. In the following reaction, the major product is



42. Match the following

List-I (aquated ion)		List-II (colour)	
A)	Ni^{2+}	I)	violet
B)	Fe^{3+}	II)	blue
C)	Mn^{3+}	III)	yellow
D)	V^{4+}	IV)	red
		V)	green

- A) $A - V, B - III, C - IV, D - II$ B) $A - IV, B - V, C - I, D - III$
 C) $A - I, B - III, C - IV, D - V$ D) $A - V, B - III, C - I, D - II$
43. If the crystal field splitting energy of a tetrahedral complex (Δ_t) of the type $[\text{ML}_4]^{n+}$ is $x\text{eV}$, what is the crystal field splitting energy with respect to an octahedral complex, $[\text{ML}_6]^{n+}$?
- A) $\frac{9x}{4}\text{eV}$ B) $\frac{9x}{8}\text{eV}$ C) $\frac{4x}{9}\text{eV}$ D) $\frac{4x}{5}\text{eV}$
44. The paramagnetic complex ion which has no unpaired electrons in t_{2g} orbitals is
- A) $[\text{Fe}(\text{CN})_6]^{4-}$ B) $[\text{Fe}(\text{CN})_6]^{3-}$ C) $[\text{Zn}(\text{NH}_3)_6]^{2+}$ D) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
45. Consider the following sequence of reactions



The incorrect statement about z is

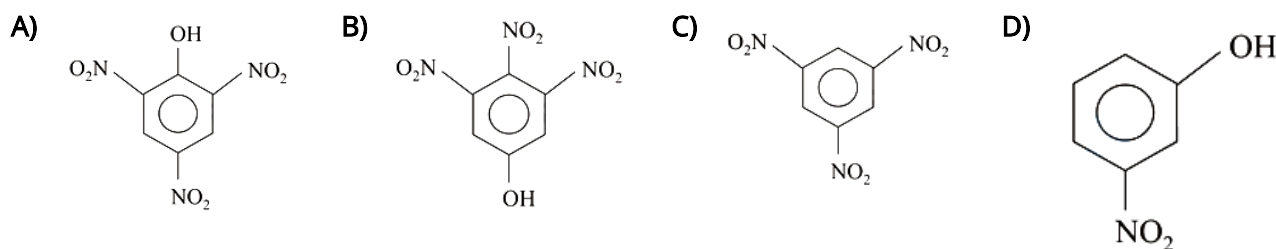
- A) z gives yellow precipitate of CHI_3 with $\text{NaOH} + \text{I}_2$ solution

B) z gives isopropyl alcohol on reduction with H_2 in the presence of Pd catalyst

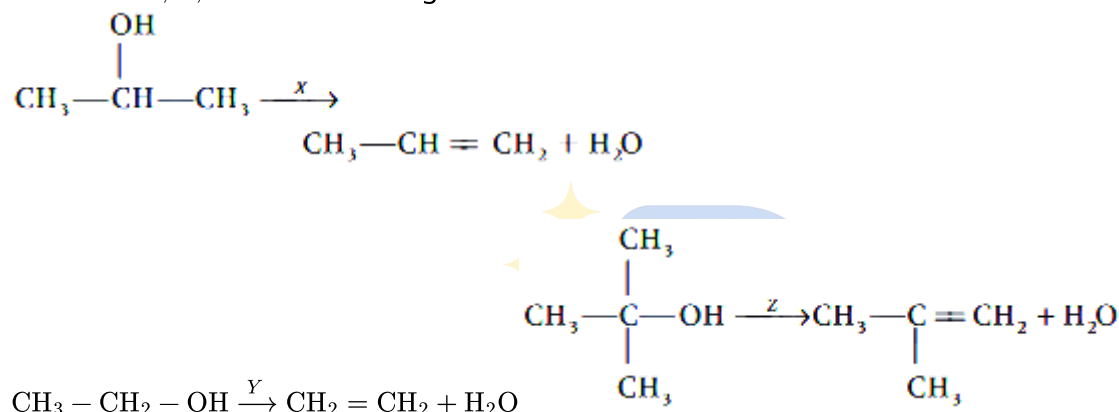
C) z on reaction with CH_3MgBr followed by hydrolysis gives 2° alcohol

D) z does not give positive test with Fehling's reagent

46. An organic compound A (C_6H_7N) on reaction with $NaNO_2 / HCl$ at $273 - 278\text{ K}$ followed by warming with water gave B. B reacts with concentration HNO_3 to give C. What is C?



47. What are X, Y, Z in the following reactions?



A)

X	Y	Z
$H_2SO_4, 443\text{ K}$	$85\%H_3PO_4, 440\text{ K}$	$20\%H_3PO_4, 358\text{ K}$

B)

X	Y	Z
$85\%H_3PO_4, 440\text{ K}$	$H_2SO_4, 443\text{ K}$	$20\%H_3PO_4, 358\text{ K}$

C)

X	Y	Z
$20\%H_3PO_4, 358\text{ K}$	$H_2SO_4, 443\text{ K}$	$85\%H_3PO_4, 440\text{ K}$

D)

X	Y	Z
$H_2SO_4, 443\text{ K}$	$20\%H_3PO_4, 358\text{ K}$	$85\%H_3PO_4, 440\text{ K}$

48. Calculate the molar ratio of a weak acid HA ($K_a = 10^{-6}$) and its salt with strong base, so that the pH of buffer solution is 6.

A) 10 B) 1 C) 6 D) 0.1

49. The solubility products of NiS, ZnS, CdS and HgS are $4.7 \times 10^{-5}, 1.6 \times 10^{-24}, 8 \times 10^{-27}$ and 4×10^{-53} respectively. An aqueous solution contains $Ni^{2+}, Zn^{2+}, Cd^{2+}$ and Hg^{2+} of equal concentration. H_2S gas was passed into this solution very slowly. The first and the last ions that precipitate as sulphides are respectively.

A) Ni^{2+}, Hg^{2+} B) Hg^{2+}, Cd^{2+} C) Zn^{2+}, Hg^{2+} D) Hg^{2+}, Ni^{2+}

50. A hydrocarbon containing C and H has 92.3%C. When 39 g of hydrocarbon was completely burnt in O_2 , x moles of water and y moles of CO_2 were formed. x moles of water is sufficient to liberate 0.75 moles of H_2 with Na metal. What is the weight (in g) of oxygen consumed?
(C = 12u; H = 1u)
- A) 120 B) 240 C) 360 D) 480

Maths - Section A (MCQ.)

51. If expression $x + \frac{1}{x^2}$, $|x| > 0$ attains its minimum value at $x = \alpha$, then α^3 is
- A) 2 B) 3 C) 4 D) 5
52. If α, β and γ are in AP and $\tan^{-1} \alpha, \tan^{-1} \beta$ and $\tan^{-1} \gamma$ are also in AP, then
- A) $\alpha - \beta - \gamma = 0$ B) $\alpha = \beta = \gamma$
C) $\alpha + \beta = \gamma$ D) $2\alpha = 3\beta = \gamma$
53. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n+1}$ is equal to
- A) e^{-1} B) $\log 2 - 1$ C) 1 D) 0
54. If $\log_3 2, \log_3 (2^x - 5)$ and $\log_3 (2^x - \frac{7}{2})$ are in AP, the value of x is
- A) 2 B) 3 C) 0 D) $\frac{1}{3}$
55. The differential equation of all circles passing through the origin and having their centre on the X -axis is
- A) $x^2 = y^2 + xy \frac{dy}{dx}$ B) $x^2 = y^2 + 3xy \frac{dy}{dx}$ C) $y^2 = x^2 + 2xy \frac{dy}{dx}$ D) $y^2 = x^2 - 2xy \frac{dy}{dx}$
56. If $a > 2b > 0$, then positive value of m for which $y = mx - b\sqrt{1+m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x-a)^2 + y^2 = b^2$ is
- A) $\frac{2b}{\sqrt{a^2-4b^2}}$ B) $\frac{\sqrt{a^2-4b^2}}{2b}$ C) $\frac{2b}{a-2b}$ D) $\frac{b}{a-2b}$
57. If $(m_i, \frac{1}{m_i})$ are four distinct points on a circle, then
- A) $m_1 m_2 m_3 m_4 = 1$ B) $m_1 m_2 m_3 m_4 = -1$
C) $m_1 m_2 m_3 m_4 = \frac{1}{2}$ D) $\frac{1}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} + \frac{1}{m_4} = \frac{1}{4}$
58. If the coefficients of x^3 and x^4 in the expansion of $(1 + ax + bx^2)(1 - 2x)^{18}$ in powers of x are both zero, then (a, b) is equal to
- A) $(14, \frac{272}{3})$ B) $(16, \frac{272}{3})$ C) $(16, \frac{251}{3})$ D) $(14, \frac{251}{3})$
59. If the number of terms in the expansion of $(x\sqrt{180} + \sqrt[3]{432})^{200}$ having integral coefficients is n , then the value of $[\frac{n}{6}]$ is (where $[.]$ denotes greater integer function)
- A) 4 B) 5 C) 6 D) 7
60. Let ω is an imaginary cube roots of unity, then the value of $2(1 + \omega)(1 + \omega^2) + 3(2\omega + 1)(2\omega^2 + 1) + \dots + (n+1)(n\omega + 1)(n\omega^2 + 1)$ is
- A) $[\frac{n(n+1)}{2}]^2 + n$ B) $[\frac{n^2(n+1)^2}{4}]$ C) $[\frac{n(n+1)}{2}]^2 - n$ D) None of these
61. If for all $x, y \in N$, there exists a function $f(x)$ satisfying $f(x + y) = f(x) \cdot f(y)$ such that $f(1) = 3$ and $\sum_{x=1}^n f(x) = 120$, then value of n will be
- A) 4 B) 5 C) 6 D) None of these

62. The distance of the point $(3, 8, 2)$ from the line $\frac{x-1}{2} = \frac{y-3}{4} = \frac{z-2}{3}$ measured parallel to the plane $3x + 2y - 2z + 15 = 0$, is
- A) 2 B) 6 C) 9 D) 7
63. $f(x) = \begin{cases} 2a - x & \text{in } -a < x < a \\ 3x - 2a & \text{in } a \leq x \end{cases}$
Then, which of the following is true ?
- A) $f(x)$ is discontinuous at $x = a$ B) $f(x)$ is not differentiable at $x = a$
C) $f(x)$ is differentiable at $x \geq a$ D) $f(x)$ is continuous at all $x < a$
64. If a discrete random variable X has the probability distribution $P(X = x) = k \frac{2^{2x+1}}{(2x+1)!}$, $x = 0, 1, 2 \dots \infty$, then $k =$
- A) $\sinh 2$ B) $\sec 2$ C) $\operatorname{cosech} 2$ D) $\cosh 2$
65. The number of persons joining a cinema ticket counter in a minute follows a Poisson distribution with parameter 6, then the probability that at least one and at most five persons join the queue in a particular minute is
- A) $e^{-6 \times 6}(25.48)$ B) $e^{-6} \left(\frac{6}{2} + \frac{6^2}{2!} + \frac{6^3}{3!} \right)$
C) $6 \times e^{-6}(29.8)$ D) $e^{-6} \left(6 + \frac{6^2}{2} + \frac{6^3}{3!} + \frac{6^4}{4!} \right)$
66. Events A, B and C are mutually exclusive events such that $P(A) = \frac{3x+1}{3}$, $P(B) = \frac{1-x}{4}$ and $P(C) = \frac{1-2x}{2}$. The set of possible values of x are in the interval
- A) $\left[\frac{1}{3}, \frac{1}{2} \right]$ B) $\left[\frac{1}{3}, \frac{2}{3} \right]$ C) $\left[\frac{1}{3}, \frac{13}{3} \right]$ D) $[0, 1]$
67. A box contains 30 toys of same size in which 10 toys are white and all the remaining toys are blue. A toy is drawn at random from the box and it is replaced in the box after noting down its colour. If 5 toys are drawn in this way, then the probability of getting atmost 2 white toys is
- A) $\left(\frac{6}{9} \right)^2$ B) $\left(\frac{8}{9} \right)^2$ C) $\left(\frac{7}{9} \right)^2$ D) $\left(\frac{2}{3} \right)^5$
68. In a city, 10 accidents take place in a span of 50 days. Assuming that the number of accidents follow the Poisson distribution, the probability that three or more accidents occur in a day, is
- A) $\sum_{k=3}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!}$, $\lambda = 0.2$ B) $\sum_{k=3}^{\infty} \frac{e^{\lambda} \lambda^k}{k!}$, $\lambda = 0.2$
C) $1 - \sum_{k=0}^3 \frac{e^{-\lambda} \lambda^k}{k!}$, $\lambda = 0.2$ D) $\sum_{k=0}^3 \frac{e^{-\lambda} \lambda^k}{k!}$, $\lambda = 0.2$
69. Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$ be such that $x = 0$ is the only real root of $P^1(x) = 0$. If $P(-1) < P(1)$, then in the interval $[-1, 1]$
- A) $P(-1)$ is not minimum of $P(x)$, but $P(1)$ is the maximum of $P(x)$
B) $P(-1)$ is minimum of $P(x)$, but $P(1)$ is not the maximum of $P(x)$
C) Neither $P(-1)$ is the minimum nor $P(1)$ is the maximum of $P(x)$
D) $P(-1)$ is the minimum and $P(1)$ is the maximum of $P(x)$
70. $A(1, 15), B(3, -12), C(6, 12)$ are three consecutive turning points of a continuous curve $y = f(x)$. If $f(x) = 0$ only for $x = \alpha$ and $x = \beta$, then $|\beta - \alpha| <$
- A) 27 B) 2 C) 5 D) 25
71. If the straight line $x \cos \alpha + y \sin \alpha = p$ touches the curve $\left(\frac{x}{a} \right)^n + \left(\frac{y}{b} \right)^n = 2$ at the point (a, b) on it and $\frac{1}{a^2} + \frac{1}{b^2} = \frac{k}{p^2}$ then $k =$
- A) 4 B) 5 C) 6 D) 7

72. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $|f(x) - f(4)| \leq 2|x - 4|^{\frac{3}{2}} \forall x, y \in \mathbb{R}$. If $f(0) = 1$, then $\int_0^1 f^2(x) dx =$
- A) -2 B) $\frac{1}{2}$ C) 0 D) 1
73. Number of triangles whose vertices are the points (x, y) in the XY-plane with integer coordinates satisfying $0 \leq x \leq 4$ and $0 \leq y \leq 4$ is
- A) 2300 B) 2260 C) 2160 D) 2230
74. p points are chosen on each of the three coplanar lines. The maximum number of triangles formed with vertices at these points is
- A) $p^3 + 3p^2$ B) $\frac{1}{2}(p^3 + p)$ C) $\frac{p^2}{2}(5p - 3)$ D) $p^2(4p - 3)$

75.

$x_1, x_2, \dots, x_n$ are n observations with mean $\bar{x}$ and standard deviation σ . Match the items of List-I with those of List-II

	List - I		List - II
(a)	$\sum_{i=1}^n (x_i - \bar{x})$	(i)	Median
(b)	Variance (σ^2)	(ii)	Coefficient of variation
(c)	Mean deviation	(iii)	Zero
(d)	Measure used to find the homogeneity of given two series	(iv)	Mean of the absolute deviations from any measure of central tendency
		(v)	Mean of the squares of the deviations from mean

- A) a b c d
(i) (ii) (iii) (iv)
- C) a b c d
(iii) (v) (iv) (ii)

- B) a b c d
(i) (iv) (iii) (ii)
- D) a b c d
(iii) (v) (ii) (i)